Inventory Management System Pro

Final Database Schema - MySQL 8 / InnoDB / utf8mb4 - Updated 05.09.2026

Scope

This document is the final database reference for InventoryPro. It includes the customer and invoice modules, invoice line items, and the automatic link between sales invoices and stock-out transactions.

Data model at a glance

Identity	users	Accounts, password hashes, and admin/employee roles
Catalog	categories, suppliers, products	Product master data, availability, price, and stock
Customers	customers	Billing address, communication details, and active status
Sales	invoices, invoice_items	Invoice header, status, totals, products, quantities, and captured prices
Inventory	inventory_transactions	Auditable stock changes with type, reason, reference, and user

Core integrity rules

• All tables use InnoDB. All relationships use ON UPDATE CASCADE and ON DELETE RESTRICT.

• Product price, product stock, invoice total, and captured unit price cannot be negative.

• Invoice and inventory quantities must be greater than zero.

• Each invoice number is unique; each product can occur only once per invoice.

• A stock transaction with reason other requires a non-empty note.

Master and identity tables

categories

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
name	VARCHAR(100)	UNIQUE, NOT NULL
description	TEXT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

users

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
username	VARCHAR(50)	UNIQUE, NOT NULL
email	VARCHAR(255)	UNIQUE, NOT NULL
password_hash	VARCHAR(255)	NOT NULL
role	ENUM(admin, employee)	NOT NULL, DEFAULT employee
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

suppliers

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
company_name	VARCHAR(150)	UNIQUE, NOT NULL
contact_person	VARCHAR(100)	NULL
email	VARCHAR(255)	NULL
phone	VARCHAR(30)	NULL
street	VARCHAR(150)	NULL
house_number	VARCHAR(20)	NULL
postal_code	VARCHAR(20)	NULL
city	VARCHAR(100)	NULL
country	VARCHAR(100)	NULL
description	TEXT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

Customer and product tables

customers

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
customer_name	VARCHAR(150)	NOT NULL
contact_person	VARCHAR(100)	NULL
email	VARCHAR(255)	NULL
phone	VARCHAR(30)	NULL
street	VARCHAR(150)	NOT NULL
house_number	VARCHAR(20)	NOT NULL
postal_code	VARCHAR(20)	NOT NULL
city	VARCHAR(100)	NOT NULL
country	VARCHAR(100)	NOT NULL
description	TEXT	NULL
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

products

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
name	VARCHAR(150)	NOT NULL
sku	VARCHAR(100)	UNIQUE, NOT NULL
price	DECIMAL(10,2)	NOT NULL, CHECK >= 0
stock_quantity	INT	NOT NULL, CHECK >= 0
is_active	BOOLEAN	NOT NULL, DEFAULT TRUE
category_id	INT	FK -> categories.id, NOT NULL
supplier_id	INT	FK -> suppliers.id, NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

Invoice tables

Invoice totals use the captured unit_price from invoice_items. The current product price can change later without changing historical invoices.

invoices

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
invoice_number	VARCHAR(50)	UNIQUE, NOT NULL
customer_id	INT	FK -> customers.id, NOT NULL
user_id	INT	FK -> users.id, NOT NULL
invoice_date	DATE	NOT NULL, DEFAULT CURRENT_DATE
status	ENUM(open, paid)	NOT NULL, DEFAULT open
total_amount	DECIMAL(10,2)	NOT NULL, CHECK >= 0
note	TEXT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

invoice_items

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
invoice_id	INT	FK -> invoices.id, NOT NULL
product_id	INT	FK -> products.id, NOT NULL
quantity	INT	NOT NULL, CHECK > 0
unit_price	DECIMAL(10,2)	NOT NULL, CHECK >= 0
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
UNIQUE(invoice_id, product_id)	CONSTRAINT	One product per invoice

Inventory transactions

inventory_transactions

Column	Type	Constraints / meaning
id	INT	PK, AUTO_INCREMENT
transaction_type	ENUM	stock_in | stock_out | adjustment_in | adjustment_out
product_id	INT	FK -> products.id, NOT NULL
user_id	INT	FK -> users.id, NOT NULL
quantity	INT	NOT NULL, CHECK > 0
reason	ENUM	purchase | sale | return_in | return_out | damaged_goods | lost_goods | recount_correction | manual_adjustment | other
transaction_reference	VARCHAR(100)	NOT NULL
note	TEXT	Required when reason = other
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	NULL, ON UPDATE CURRENT_TIMESTAMP

Invoice-to-stock workflow

For every saved invoice item, the application creates a stock_out transaction with reason sale and uses the invoice number as transaction_reference. The stock update locks the product row, validates the resulting quantity, and commits the transaction and stock update together.

Foreign-key relationships

categories.id	products.category_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
suppliers.id	products.supplier_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
customers.id	invoices.customer_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
users.id	invoices.user_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
invoices.id	invoice_items.invoice_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
products.id	invoice_items.product_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
products.id	inventory_transactions.product_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT
users.id	inventory_transactions.user_id	1:N	ON UPDATE CASCADE / ON DELETE RESTRICT